Pricing a new market with an AI-assisted actuarial workflow

IAA AI-Assisted Hackathon 2026. Team “The KMR”, a solo entry by a Fellow of the CIA. Report v2, 28 October 2026. Every number comes from the submitted repository; file paths are given where a reader may want the detail.

1. Summary

The task. The insurer is expanding from a developed market into an emerging one. We predict the risk premium for every 2025 policy row in three lines: private motor, home buildings and home contents. The risk premium is the expected ultimate cost of claims with a loss date in the row’s period. It is net of excess and recoveries, includes claims not yet reported or settled, and excludes declined claims. It is a total for the row, not an annual rate, and it is not the premium charged. Accuracy is scored by Tweedie D^2 (power 1.5) against a constant-rate baseline. The report also builds the office premium for 20 reference policies (section 8).

The approach. An AI agent proposes model changes, and a frozen validation harness decides which ones pass. The actuary owns everything that defines the problem: data treatment, the claims target, development, trend, the validation design and the scorer. The agent cannot edit any of these. Changes that pass the harness then go through a formal review before the actuary adopts them (section 4).

The results.

	Validation D^2	Public leaderboard	Private leaderboard
Part 1, developed market (final)	0.1425 (weighted mean of folds A and B)	0.1262	0.1336
Part 2, emerging market (selected)	0.1535 / 0.1286 (level-neutral, folds B / R)	0.1662	after close

Part 2 folds are scored level-neutral because they cannot test the 2025 level (section 7). The keep rule uses the deploy metric instead (section 6).

Key messages. Relativities estimated on the developed market carried over better than two small emerging years could estimate them. The emerging data then adjusted the level of each peril and added one rating factor, gender. Most other ideas were tested and dropped because they did not hold on both validation folds. The largest remaining uncertainty is the 2025 level, which no historical fold can test (section 7).

Term	Meaning in this report
Tweedie D^2	The share of prediction error removed compared with one constant rate for every row. Higher is better; 0 means no better than the constant
Public / private leaderboard	Kaggle scores a random third of the 2025 rows publicly during the competition; the other two-thirds decide the final ranking
Fold	One train-and-score split of the historical years. In Part 2, fold B trains on 2023 and scores 2024, and fold R trains on 2024 and scores 2023
Relativity	How much one risk costs compared with another, apart from the overall level
Level	The overall cost per unit of exposure for a line or group
Level-neutral D^2	D^2 after each line’s predictions are rescaled to its actual total, so it scores ranking only
A/E	Actual claims divided by expected (predicted) claims
Keep rule	The pre-declared test a change must pass to be kept (section 6)

2. Modelling approach

Data. AI profiling found eight data-quality issues, and the actuary confirmed each one (`reports/data_quality.md`). Two would have broken the model. Customer-service contacts

include claim calls in training but not in the test files, so the field is excluded as leakage. About 0.4% of exposures were recorded in days, so exposure is recomputed from the policy dates.

Target. Claims are valued at 31 December 2024, so recent years are immature. The training target is therefore developed to ultimate with a chain ladder on report delay, one triangle per line and peril group. An automated check requires developed 2024 to land on the 2021-23 trend. Raw motor was 12% short; developed, it is within 2%. Costs are then trended to 2025.

Part 1 model. Each line is a blend of gradient-boosted trees (LightGBM, CatBoost) and an Explainable Boosting Machine, fitted with Tweedie loss. Motor bodily injury uses a separate frequency model with a flat severity. Monotone constraints keep excesses, no-claims years and flood zone in the expected direction. The actuary excluded the incumbent's premium, email domain, sales fields and customer claim history on admissibility grounds, not on score.

Part 1 validation. Two out-of-time folds tested each change. Fold A trains on 2021-22 and scores 2023; fold B trains on 2021-23 and scores 2024. A change was kept only if it beat the best run by more than seed noise on both folds and repeated on fresh seeds.

3. Adaptation to the new market

What changed (`reports/part2_drift.md`). Emerging claims are 20-40% cheaper than developed claims at the same maturity. Motor claims are 1.2-1.8 times as frequent. Credit band, mileage and telematics are missing, ages come in bands, and gender is added. The 2025 book is 77% new business from a broker transfer, against 6-9% in the historical years.

What carried over from Part 1. The Part 1 model refitted on emerging data alone scored 0.129 on the public leaderboard. The same model fitted on the developed market and applied to emerging rows scored 0.142. Every Part 2 model therefore starts from a developed base. Development patterns and trends also carry over, because two emerging years cannot estimate them. The level method carries over too: the 2025 level comes from balancing the model on its training rows, so the 2025 business mix moves the level through the relativities.

How the relativity model was developed. The agent worked through a pre-registered list of changes, one mechanism per run. Table 1 shows what was tested and what was kept.

Change tested	Evidence	Decision
Part 1 model refitted on emerging data only	Public 0.129, against 0.142 for the developed fit	Dropped (actuary)
Developed base refitted separately for each peril (P0)	Each peril takes its own emerging level	Kept (rule)
Motor gender, an adjustment on the base kept only if a likelihood-ratio test passes (L1)	Level-neutral D^2 up on both folds; public +0.007	Kept (rule; admissibility by actuary)
Joint models with six to eight emerging factors	Extra factors diluted gender	Dropped (rule)
Eleven small factors stacked by the agent	Folds +0.007, public -0.003	Reverted (actuary; rule tightened)
Per-peril emerging levels, pooled contents base, new-business factor (O4)	Level-neutral D^2 +0.001 (B) and +0.002 (R); public +0.004	Kept (pre-registered by actuary, passed rule)
20 further mechanisms: weather, region, previous insurer, missingness, new-business regime, recalibration	Each lowered fold R or gained less than 0.002	Dropped (rule; three-model review)
Neural-network base (TabM), 25% blend	+0.0016 overall, fold B and buildings down	Dropped (rule)
Contents trend 0% and motor bodily-injury trend 12%	Both folds' A/E move towards 1	Kept (actuary, from emerging data)

Changes learned on emerging rows helped fold B a little at best and always hurt fold R. Changes that replaced the developed base hurt fold B. The selected model is O4 with the 2023-24 trends.

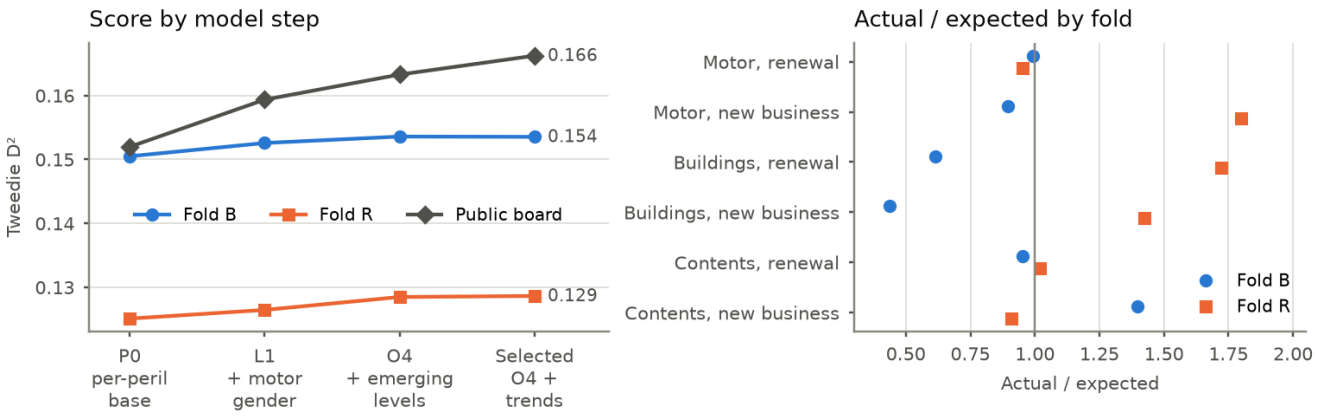


Figure 1. Left: score of each kept step on the two folds (level-neutral) and on the public leaderboard. Right: A/E of the selected model by line and new business, on each fold.

Figure 1 shows the relativities improving on both folds through O4. The last step changes the level, not the ranking, so its fold scores are flat. The right panel shows the main weakness. Buildings A/E is 0.6 on one fold and 1.7 on the other, because fire claims are few (49 in 2023, 21 in 2024). Motor new business swings from 0.89 to 1.80 for the same reason. Motor and contents renewals sit within 5% of 1 on both folds.

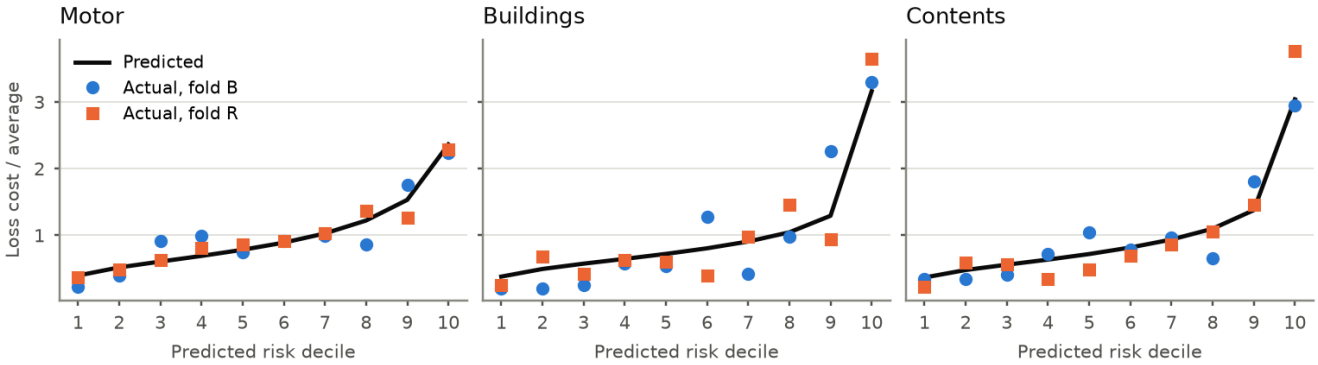


Figure 2. Lift by predicted risk decile, selected model, both folds. Each value is the decile's loss cost divided by the line average.

Figure 2 checks that the ranking holds out of time. On every line, actual cost broadly follows the predicted curve on both folds. The model predicts the top motor decile at six times the bottom, and the folds show six to ten times. The scatter in the middle buildings deciles comes from the few fire claims.

4. Major decisions and human review

The review process. The harness runs the agent's experiments in batches, and each run has a time limit. Batches continue until the agent reports the list exhausted or a batch cap is reached. A formal review follows before anything enters the model. The review uses statistical validation: fold scores, a paired bootstrap, and A/E by line, fold and segment. It also uses review plots of the kind used in traditional empirical pricing. In Part 1 the actuary reviewed a dashboard rebuilt after each of 27 batches. In Part 2 the reviews on 25 and 27 October used the fold statistics and A/E tables. Figures 1 and 2 are the final review of the selected model. Only the actuary adopts a change, and every leaderboard submission has a sign-off row (`docs/SIGNOFF.md`).

Where the review changed the method.

- On 25 October the agent kept eleven small factors, stacked in pairs that each just missed the margin. The folds gained and the leaderboard lost, because each factor was chosen on the claims

it was scored on. The actuary reverted them. The keep rule now requires a pre-registered list, one mechanism per run, a claim-count test on every factor and no stacking of rejected runs.
- An outside model built a recommendation on 2023 being all new business. The data shows 6.4%, so the recommendation was rejected.
- Reviewers found that a claim-count model offset by a cost prediction learns inverse-severity effects. The agent's correction blocks built that way were dropped. O4's new-business factor uses the same construction; a cost-based replacement did not pass, so it stays and is listed as a limitation in section 7.
- An earlier pooled base duplicated emerging rows to weight them, which let the trees memorise claims. That version was not used, and a sample-weighted rebuild was tested instead.
- Gender is admissible in the emerging market under the brief. Its use was an owner decision, recorded in the assumptions log.

5. Use of AI

Claude Code agents profiled the data, researched methods (`research/sources.md`), wrote most of the code and ran the experiment loop. In the loop, a headless agent may edit only the experiment file, and it must state a hypothesis first. The driver reverts any other change, runs the frozen harness and writes the result to the ledger (`results.tsv`, `results_emerging.tsv`). The loop logged 248 experiments: 166 in Part 1 and 82 in Part 2. Models from other providers challenged the work. On 25 October OpenAI GPT-6.1 Sol reviewed the design. On 27 October four models proposed ideas and three reviewed the results, for $2.90. Heavy refits ran on private Kaggle notebooks, including a GPU neural network.

The harness scored every agent proposal, and the agent could not change the scorer. Outside-model claims were checked against the data (section 4).

6. Guardrails and reproducibility

The keep rule was declared before any candidate was scored. In Part 2 a change must gain more than 0.002 on the deploy metric, and neither fold may fall by more than 0.003. The 10th percentile of a paired bootstrap must also be above zero. The deploy metric weights the natural book two-thirds and the book reweighted to the 2025 mix one-third. The harness, target, folds and scorer are covered by 41 automated tests, which run before every loop iteration. Pre-commit runs ruff and gitleaks, and CI repeats them with the tests. Assumptions sit in one config file per market, with the reason beside each value. The selected file can be rebuilt from the repository (`scripts/part2_candidate.py`, hash-checked). The work follows the CIA Standards of Practice for data, models, documentation, assumptions and reporting, and the IAA AI Task Force papers (`docs/governance.md`).

7. Assumptions, limitations and uncertainty

The 2025 level. The folds can judge relativities but not the 2025 level. Each fold trains on a book with about 8% new business, while 2025 is 77% new business. The public leaderboard scores a random third of the same 2025 book. Two probe submissions per group therefore give the best level for that group on the public rows. Each read was shrunk towards 1 by its sampling error, and was used only after it held on the folds under checks set in advance. Where our own data explained a read, the trend was changed instead of applying a multiplier. A real insurer would not have this read, so it is disclosed as a reliance.

Group	Leaderboard read vs O4	Used	Explained by own data
Motor new business	1.21	1.185	5% (bodily-injury trend)
Motor renewals	1.10	1.07	5% (bodily-injury trend)
Buildings	1.00	1.00	n/a
Contents	0.90	0.93	All (contents trend 0%)

Item	Effect
Case reserves on open claims taken as adequate	±10% moves the 2025 motor level ±3.4% and buildings ±1.9%
Emerging motor bodily injury reports more slowly than developed	Would raise motor ultimates 2-3%; logged as a sensitivity
New-business factor rests on thin emerging data and a count-on-cost construction	New-business A/E 0.89 and 1.80 on the folds
Buildings level rests on few fire claims	A 10% buildings level error costs 0.004-0.005 of D^2
Fold gap	Fold R scores about 0.025 below fold B; it trains on the less developed year
Weather in 2025 is milder	Developed elasticities put the effect under 1%; not modelled

Reliance and deviations. The data is synthetic and supplied by the host; it was used as given after the data-quality treatment in section 2. Neither fold contains a broker book. With more time, the first change would be a grouped multi-year validation design.

8. Office premium for the reference policies

The office premium adds expenses, commission, tax, catastrophe reinsurance and profit to the predicted risk premium, using the emerging assumptions in the `expenses` table supplied with the data:

Office premium = (risk premium + fixed expense + cat reinsurance) / (1 - variable expense - commission - premium tax - profit margin).

The fixed expense is 85 CU per policy. Cat reinsurance is 12% of the risk premium on buildings and contents. Variable expense is 15%, premium tax 8% and profit margin 8%. Commission is 20% for broker business and nil for direct. Each row below adds back to the office premium. Risk premiums come from the selected file (`scripts/part2_office_premium.py`).

Policy	Line	Channel	Exposure	Risk premium	Cat reins.	Fixed	Variable	Commission	Tax	Profit	Office premium
M00161473	motor	broker	0.964	1,446.02	0.00	85.00	468.68	624.90	249.96	249.96	3,124.52
M00176777	motor	broker	0.992	1,128.34	0.00	85.00	371.43	495.24	198.10	198.10	2,476.20
M00185311	motor	direct	0.964	969.51	0.00	85.00	229.24	0.00	122.26	122.26	1,528.28
M00167496	motor	broker	0.984	624.19	0.00	85.00	217.10	289.47	115.79	115.79	1,447.33
M00129413	motor	broker	0.332	148.36	0.00	85.00	71.44	95.25	38.10	38.10	476.24
M00118569	motor	direct	0.940	625.98	0.00	85.00	154.56	0.00	82.43	82.43	1,030.41
M00153851	motor	broker	0.962	323.09	0.00	85.00	124.93	166.57	66.63	66.63	832.83
B00229821	buildings	broker	0.992	315.87	37.90	85.00	134.32	179.09	71.64	71.64	895.45
B00218282	buildings	broker	0.973	320.99	38.52	85.00	136.07	181.43	72.57	72.57	907.16
B00250679	buildings	broker	0.110	42.44	5.09	85.00	40.57	54.10	21.64	21.64	270.48
B00223369	buildings	broker	0.995	544.67	65.36	85.00	212.77	283.69	113.48	113.48	1,418.44
B00215809	buildings	direct	0.975	449.67	53.96	85.00	127.96	0.00	68.25	68.25	853.09
B00215707	buildings	broker	0.986	303.60	36.43	85.00	130.11	173.48	69.39	69.39	867.41
B00217584	buildings	broker	0.967	866.82	104.02	85.00	323.22	430.95	172.38	172.38	2,154.77
C00259568	contents	broker	1.000	42.93	5.15	85.00	40.74	54.32	21.73	21.73	271.59
C00283470	contents	broker	0.973	63.28	7.59	85.00	47.72	63.62	25.45	25.45	318.10
C00270331	contents	broker	0.973	35.89	4.31	85.00	38.33	51.10	20.44	20.44	255.51
C00314425	contents	broker	0.069	5.03	0.60	85.00	27.75	36.99	14.80	14.80	184.97
C00302985	contents	broker	0.992	44.42	5.33	85.00	41.25	55.00	22.00	22.00	275.00
C00286287	contents	broker	0.967	110.04	13.21	85.00	63.75	85.00	34.00	34.00	425.00